

Optimizing Pharmaceutical Control with Multi-Task Contextual Bandits: Addressing Batch Heterogeneity for Improved Manufacturing Efficiency

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Problem definition: Efficient control is critical for continuous pharmaceutical manufacturing to ensure the production of high-quality tablets despite variations in raw material attributes and environmental factors. Furthermore, the data in 2024 MSOM Data-Driven Research Challenge indicates that batches with the same observed attributes are heterogeneous in their impact on the tablet quality, i.e., there is hidden batch heterogeneity. We explore how to design optimal pharmaceutical control policies with limited data and hidden batch heterogeneity. **Methodology/Results:** We propose a contextual bandit model with related sequential batches to control a manufacturing process where the raw material batches have hidden heterogeneity. We propose a novel algorithm that combines results from the multi-task learning and the contextual bandit literature to minimize the regret in this setting. We show that the batch heterogeneity is best represented by a new multi-task random forest model. We extend the theoretical regret guarantees for both the Thompson Sampling and greedy algorithms to the setting of contextual bandits with related sequential batches. The results of our numerical experiments show that the multi-task random forest has good prediction performance, and the control policy recommended by our algorithmic framework outperforms the historical policy by as much as 20%. **Managerial implications:** In practice, overlooking hidden heterogeneity between batches can lead to a significant loss in quality. Our framework is flexible to model the production process with limited but relevant data, offering a valuable tool for improving pharmaceutical manufacturing and related production problems.

Key words : Multi-Task Learning, Contextual Bandit, Data Heterogeneity, Pharmaceutical Production

1. Introduction

In pharmaceutical production, continuous manufacturing provides significant advantages over traditional batch processing. It enables real-time monitoring and control, allowing manufacturers to adjust parameters dynamically in response to environmental and material variations and improving product consistency and quality (Domokos et al. 2021). This approach also enhances efficiency and cost-effectiveness by reducing downtime and ensuring a smooth production flow with real-time signals and sensors. Precisely controlling manufacturing processes is essential to adhere to the regulatory standards set by governing

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that accounting for heterogeneity is very important for out-of-sample accuracy. Ignoring heterogeneity leads to severe degradation in prediction performance, with the R^2 dropping from 0.15 to negative. The training details are in Section 4.2.

0.15 0.10 0.05 0.00

-0.05 -0.10

Figure 1

Pool 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 Batch ID

Out-of-batch performance of the prediction model for the tablet quality cost that does not take batch heterogeneity into account for the first 40 batches ($ID \leq 40$), where “Pool” means the out-of-sample performance within these 40 batches and the others refer to the IDs of batches. R^2 can be negative since the model transfers poorly to batches that are not in the training data. Here we set the beginning of each batch to the epoch when the active pharmaceutical ingredient (API) is refilled.

This result in Figure 1 implies that the prediction model needs to be adapted to each new batch by interacting with its data. In this paper, we propose a new prediction model that starts with a baseline estimate for each batch and then progressively refines it as more data from the batch becomes available. Building on this, we introduce a new decision framework that we call contextual bandit with related sequential batches to choose actions. Our results demonstrate that accounting for batch heterogeneity enhances predictive accuracy (see Section 6.2), and that the actions selected by the proposed bandit model lead to lower incurred costs (see Section 6.3). We build on multi-task learning (MTL) (Baxter 2000), more specifically, the multi-task bandit literature, which proposes general algorithms to transfer disinformation between related problem instances (tasks) to improve predictive or prescriptive analytics. However, results in the literature either study the setup

where each task is learned simultaneously (Yang et al. 2021, Duan and Wang 2023) instead of sequentially, as is the case in our setting, or under the linear model assumption (i.e., the setup in Bastani et al. (2022)). Our work bridges these gaps and proposes the contextual bandit with related sequential batches (MTL-Seq-CB) model, and an algorithmic framework for

The rest of this paper is organized as follows. In Section 2 we review the related literature. In Section 3 we discuss the results of our data analysis that clearly show the existence of hidden batch heterogeneity. We describe the Seq-CB model in Section 4 and present bounds for our proposed MTL-Seq-CB framework in Section 5. In Section 6 we discuss the results of our numerical experiments, and the associated insights. In Section 7 we include some concluding remarks. Proofs and additional results can be found in the online appendix.

We use notations O , Ω and $\tilde{O}$ to describe the asymptotic performance of a function. Specifically, given two functions $f(T)$ and $g(T)$, $g(T)$ is said to be $O(f(T))$ (and $f(T) = \Omega(g(T))$) if there exist positive constants c , and T_0 such that $g(T) \leq cf(T)$ for all $T \geq T_0$. Similarly, $g(T)$ is said to be $\tilde{O}(f(T))$ if $g(T)$ is $O(f(T) \cdot \log k T)$ for some constant k .

2. Literature Review

2.1. Chemical Dynamics in Manufacturing Control

Conventional control methods for pharmaceutical manufacturing identify control policies for action selection on the basis of analytical models for the chemical and mechanical dynamics of the processing system as a function of the process parameters, environmental variables, e.g., temperature and humidity, and raw materials attributes. For example, in blending, the action is the screw speed (Kumar et al. 2014) that affects the uniformity of the downstream resulting flow; and in compression, the action is the compression force (Diab et al. 2021) that impact the physical properties of the tablets (Moghtadernejad et al. 2018). Note that both these actions have nonlinear effects (Adeleye 2019). Among environmental factors, temperature and humidity have the largest impact on tablet quality (Prado and Vajta 2021). Moisture has been shown to affect the flowability of pharmaceutical excipients (Crouter and Briens 2014), adding unpredictability and uncertainty to the manufacturing process. The physicochemical properties of the ingredients explain variabilities in the production process, e.g., bulk density (Kreiser et al. 2023), order of mixing ingredients (Pingali et al. 2011). Despite these qualitative insights, the exact dynamics of each production pipeline are often unique and analytically challenging to identify. This

underscores the need for data-driven operational techniques to analyze and optimize manufacturing processes on a case-by-case basis (Gams et al. 2014).

There is some existing work on integrating domain knowledge of chemical engineering applications with data-driven learning approaches for interpretable, reliable, and scalable decision-making (Venkatasubramanian 2019). Specifically, data-driven operations (Mišić

tasks (Finn et al. 2017). In the bandit setting, the meta-Thompson Sampling algorithm uses empirical Bayes to set the Gaussian prior for new tasks (Nabi et al. 2022, Kveton et al. 2021). Variants of this approach have also been applied to dynamic pricing (Bastani et al. 2022). These problem setups are closely related to ours; however, in these setups the shared representation is known, and that significantly simplifies the problem.

The effectiveness of meta-learning depends on learning an embedding for transferring information between tasks. The goal of multi-task learning (Baxter 2000) is to improve generalization and performance across multiple related tasks when the task embedding is parametric. In the supervised learning setting (Maurer et al. 2016), by assuming the unknown task parameters are close in some metric (Bastani 2021), multi-task learning improves generalization by pooling the data for all tasks and then regularizing the task parameters (Tripuraneni et al. 2020, Du et al. 2021, Duan and Wang 2023). In the bandit setup, some works investigate the problem of minimizing the total regret in multiple parallel related bandit instances (Yang et al. 2021, Hu et al. 2021, Xu and Bastani 2025). In contrast, in our setting, the tasks or bandit instances arrive sequentially.

In other operational applications, there is work on improving the prediction performance for new products and services with limited data by leveraging historical data from other related products and services. This approach has been applied to ride-sharing (Qin et al. 2020), product sales forecasting (Hu et al. 2019, Baardman et al. 2018), and inventory control (Feng et al. 2023). We contribute to this emerging area by developing a model extending this meta-learning approach to continuous pharmaceutical production.

3. Exploratory Data Analysis

In this section, we describe the key process and challenges using the data analysis from 2024 MSOM Data-Driven Research Challenge.

3.1. Summary Description of Pharmaceutical Process

We focus on the data provided by Holman et al. (2021) and synthesized by Martagan et al. (2024). The data is from a pharmaceutical production process where real-time monitoring of process parameters is provided by a GEA CDC-50 control system. The system processes

6,000kg of raw materials and manufactures over 15 million tablets over a period of 120 hours. The production process feeds all the raw materials into a blender using loss-in-weight (LiW) feeders that control the total amount and mixture of flows by measuring

Figure 2

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Summary of the pharmaceutical production process, highlighting the blending and compression stages.

Cost Flow Loss (of two blenders)

State

Cost Deviation of Hardness

Formulation Blending Action

Coating

Raw Material Attributes

Environmental Variables

We assume that the blending and compression stages can be modeled separately. Although these two stages might appear dependent, the operational range is broad enough to prevent significant coupling events. This separation assumption is supported by recent pharmaceutical literature (Pawar et al. 2016, Jakubowska and Ciepluch 2021). To empirically validate the separation assumption, we analyze the correlation between residuals of the “total flow loss” c_1 in the blending stage and the “deviation of hardness” c_2 in the compression stage after conditioning on shared state variables s . The results in Table 1 indicate that the residual costs, $c_1 - E[c_1|s]$ for blending and $c_2 - E[c_2|s]$ for compression, exhibit small correlations (< 0.1). The correlations remain low (absolute values less than 0.15) even when we use other metrics for the costs c_1, c_2 .

Table 1 Correlation between outcomes in the blending and compression stages conditioned on state variables, given by $\text{Corr}(c_1 - E[c_1|s], c_2 - E[c_2|s])$, where $E[c_1|s], E[c_2|s]$ are approximated using a random forest model. The boldfaced entries highlight the costs used in the subsequent analysis and their correlations.

Compression c_2 \ Blending c_1 Tablet Uniformity

Concentration for Blender i, and “B-i total loss” as its corresponding multiplication with the mass flow amount, “Sum Total Loss” denoting the B-1 + B-2 total loss.

3.2. Interplay between States, Actions, and Costs

Next, we evaluate the impact of state variables, including raw material attributes, on the actions and costs in blending and compression stages. As a first step, we preprocess the

B-1 Loss	B-1 Flow Loss	B-2 Loss	B-2 Flow Loss	Total Flow Loss
0.115	-0.053	0.038	0.064	-0.094
-0.114	0.073	-0.046	-0.123	0.070

Notes. We denote the “B-i loss” as the summation of the deviation of blend potency and OOS

Deviation of Hardness

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e-companion to : ec1

Online Appendix to
Manufacturing Pharmaceutical Control

EC.1. Additional Results in Section 3

Interpretation of Individual Shapley value. For an individual data point $X = (X(1), \dots, X(dx))^T$, iii the Shapley value of j -th feature $X(j)$, $j \in [d]$ represents how much that feature contributes to ix

shifting the prediction from the average prediction across all data points (Lundberg and Lee 2017). Mathematically: $f(X_i) - E[f(X)]$ is the total deviation from the mean prediction and the deviation is distributed among features. The Shapley value for the j -th feature for X_i is:

$$\phi_j = \frac{1}{dx} \sum_{S: j \in S} \frac{|S|!(dx - |S|)!}{|S|!(dx - |S|)!} [f_{S \cup \{j\}}(X_i) - f_S(X_i)], j \in [dx], S: S \subset [dx] \setminus \{j\}$$

where dx is the number of features and S is a subset of features excluding the i -th one, f_S and $f_{S \cup \{i\}}$ are the model prediction using only features in S and $S \cup \{i\}$, which is a random forest with the subset of features.

In Figure EC.1, we report the Shapley value attribution results for other actions and find both environmental variables and raw material attributes lead to significant impacts on the action.

Heterogeneity. To demonstrate the heterogeneity of costs, we consider $K = 40$, i.e., batches in fitting the random forest, and visualize the 3D projection of $\{\theta(g)\}$ of them for blending and

$i \in [K]$

compression stages in Figure EC.2. We observe a wide range of differences in each stage.

EC.2. Proofs of Theorem 1 in Section 5

EC.2.1. Detailed Results in Section 5.1.1

Across the proofs, we define the sub-Gaussian norm of $X \in \mathbb{R}^d$ as:

$$\|X\|_{\psi_2} = \inf\{c > 0: E[\exp(\|X\|^2/c^2)] \leq 2\}.$$

The following matrix concentration inequality follows from standard results in Tropp (2011):

Lemma EC.1 (Matrix Concentration Inequality). For any $x_1, \dots, x_n$ are i.i.d. random vectors in $\mathbb{R}^p$ with $\|x_i\|_2 \leq R$ (or with sub-Gaussian norm R), and $\Sigma = E[x_1 x_1^T]$ and $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$. Then there exists $C > 0$, such that for all $\delta > 0$, with probability at least $1 - \delta$:

$$\|\hat{\Sigma} - \Sigma\|_{op} \leq CR^2$$

Based on Lemma EC.1, we provide the following concentration properties of the parameters of

interest.

Lemma EC.2 (Concentration Properties for Parameters). Suppose Assumption 4 holds. Then $\forall k \in [K]$, with probability $1 - e^{-t}$ (where $t < 1$), we have: $\| \sum_{j=1}^{M+k-1} \theta_j \theta_j^T \| \geq c_0$ for some

$c_0 > 0$.

Furthermore, we have: $\sigma_{\min}(\Theta^{M+k-1}) \geq \alpha > 0$, where $\Theta^{M+k-1} = [\theta_1 | \dots | \theta_{M+k-1}] \in \mathbb{R}^{p \times (M+k-1)}$ and $\sigma_{\min}(\cdot)$ denotes the smallest positive singular value of a non-zero matrix.

Proof of Lemma EC.2. Denote $S = \sum_{j=1}^{M+k-1} \theta_j \theta_j^T$. From Assumption 4, the sub-Gaussian

norm of $\|\theta - \mu\|_2$ is bounded by $\sqrt{p \lambda_{\max}(\Sigma)}$. Then from standard random matrix concentration

inequalities in Lemma EC.1, with probability at least $1 - e^{-t}$, we have:

$\|S - \mathbb{E}[S]\|_{\text{op}} \leq \sqrt{p \lambda_{\max}(\Sigma)} \sqrt{M+k-1} \leq C \sqrt{p \lambda_{\min}(\Sigma)} \sqrt{M+k-1}$.

ec2

e-companion to :

Temperature Humidity API MS Avicel_LOT MS_LOT0 MS_LOT1 MS_LOT2

Temperature Humidity API MS Avicel_LOT MS_LOT0 MS_LOT1 MS_LOT2

Temperature Humidity API MS Avicel_LOT MS_LOT0 MS_LOT1 MS_LOT2

Figure EC.1

-2.5 -2.0

-1.5 -1.0 -0.5 0.0 0.5

SHAP value

Low

(a) Screw RPM PD2

(c) Screw RPM PD4

SHAP Beeswarm Plot for Screw RPM PD7 (R2 = 0.74)

-0.75 -0.50 -0.25 0.00 0.25 0.50 0.75 1.00

SHAP value

(e) Screw RPM PD7

(d) Screw RPM PD5

SHAP Beeswarm Plot for Main compression force (R2 = 0.87) High

(g) Main Compression Force

(h) Ejection Force Tablet

High

Low

SHAP Beeswarm Plot for Pre-compression force (R2 = 0.28)

Shapley value distribution plots of actions given state variables.

Furthermore, note that $E[S] = \Sigma + \mu\mu^T$, the minimum eigenvalue satisfies: $\rho_{\min}(S) \geq \lambda_{\min}(\Sigma) \geq 1 - M + k - 1 \geq 2l$.

e-companion to :

ec3

Figure EC.2

3D Projection visualization of heterogeneous effects of the random forest model to costs $\{\theta(g)\}_i$

(a) Blending

(b) Compression

Task-Specific Parameter Vectors of Tablet_Press Cost Projected in 3D

Multiplying the above inequality by p , we obtain:

$$p \sum_{j=1}^{M+k-1} \|\theta_j\|^2 \geq p \sum_{j=1}^{M+k-1} \|\theta_j\|^2$$

$$\sum_{j=1}^{M+k-1} \|\theta_j\|^2 \geq \sum_{j=1}^{M+k-1} \|\theta_j\|^2$$

$j=1$

For a Gaussian matrix $\Theta \in \mathbb{R}^{p \times (M+k-1)}$, recall Theorem 5.39 in Vershynin (2010) when

we consider the matrix $\Theta = [\theta_1 | \dots | \theta_{M+k-1}] \in \mathbb{R}^{p \times (M+k-1)}$ with each $\theta_i \sim N(\mu, I)$. Then with probability $1 - \exp(-t)$, the smallest singular value of a centered Gaussian matrix is bounded by:

$\sigma_{\min}(\Theta) \geq \sqrt{M+k-1} - C\sqrt{p-t}$.

Comparing in our case, each column of Θ is drawn from $N(\mu, \Sigma)$ with $\lambda(\Sigma) \geq \alpha$, then

$\sqrt{\alpha}$

In the following, for each k , we define $G := [G_1 | \dots | G_{M+k-1}] \in \mathbb{R}^{d \times (M+k-1)}$ with G_j being the

this results in: $\sigma_{\min}(\Theta) \geq$

gradient of the true loss in the j -th task. That is:

-2

0.5

-1.0

$0.023 - 0.5$

PC2

37

39

35

2247

4134

30

$M+k-1$ min
 $M+k-1-C\sqrt{p-t}\alpha\geq\alpha$, when $t<1$ and $M\geq p+1$.

$1 \times T$

$\varepsilon_j, t\phi(x_j, t, s_j, t) \in \mathbb{R}^d$. (EC.1) We can represent $EM+k-1$ with G and G_j , which follows from a restatement of Theorem A.5 from Duan

$G_j = T$

Lemma EC.3 (Estimation Error). Suppose B is unknown and Assumptions 3 and 4 hold. Let

$t=1$

and Wang (2023):

$P^* \in \mathbb{R}^{d \times d}$ be the projection onto $\text{Range}(B)$. For each $k \in [K]$, let $(B^{\wedge}, \{\theta^{\wedge}\})$ denote the

$k-1 \ j \in [M+k-1]$

solution of (9) in Algorithm 1 with the regularization parameter $\lambda > C M \max \|G\|$ for

$p\alpha^2 \ j \in [M+k-1] \ j^2$ some constant $C > 0$ in the optimization problem. Then the following bound holds, $\forall j \in [M+k-1]$:

where H is defined in Proposition 2.

$\|B^{\wedge} \theta^{\wedge} - B\theta\| \leq C \ k-1 \ j \ j^2$

$2, \alpha$

$\sqrt{}$

T. Then $\{G_i, i \in [M]\}$ is sub-Gaussian with the sub-Gaussian norm

$\frac{1}{2} \max$

$\|P^*G\| \ j \in [M+k-1] \ j^2 +$

$\sqrt{p}\|G\| \frac{1}{2}$

$\lambda_{\min}(H)$

Lemma EC.4 (Sub-Gaussian Property of the Tail loss). Suppose Assumptions 2, 1, 3, 4

and 5 hold. Define $W = \sigma D /$

being W . Conditioned on F

(including the historic data D and k new batches), then $GM+k$ is sub-Gaussian with the subGaussian norm being W .

k

($k \geq 0$), where F

k

denotes the realization of $\{(s$

i, t

, x

i, t

, $c\} \}_{M+k, T}^{i, t, t=1}$

ec4 e-companion to :

Proof of Lemma EC.4. For $i \in [M]$, the sub-Gaussian property of G_i directly follows from the fact that $\|\phi(x, s)\|_2 \leq D < \infty$ in Assumption 3. Conditioned on F_{0V}^{i-M} , $\|\varepsilon_{j,t}\phi(x_{j,t}, s_{j,t})\|_2 \leq \sigma D$. Then the sub-Gaussian property holds for $i \geq M + k$. $\square$

Lemma EC.5 (First-order Estimator Error). Suppose Assumptions 2, 1, 3 and 4 hold. Following the notation in Lemma EC.3, for any $k \in [K]$, Then there are constants C_1 and C_2 such that the following hold with probability at least $1 - \exp(-t)$:

- $\max_{j \in [M+k-1]} \|P^* G_j\|_2 \leq C_1 \sigma \sqrt{p + \log(M+k-1) + t}$;

$$\sqrt{\sqrt{\sqrt{\cdot}}}$$

$$T \cdot \|G\| \leq C_2 \sigma (d + M + k - 1 + t).$$

$$\sqrt{\cdot}$$

$$2$$

Note that when $k = 1$, the data fully relies on the offline data and becomes Theorem 5.39 in Vershynin (2010). Through Lemma EC.4, we generalize the result to the problem setup with both online and offline data, i.e., $k > 1$.

Proof of Lemma EC.5. Due to projection, we have: $\|P^* G_j\|_\psi \leq \|G_j\|_\psi \leq \sigma D q_1$. By applying 22T

Theorem 2.1 in Hsu et al. (2012b), we have:

$$rp + \log(M + k - 1) + t!$$

$$P \max \|P^* G_j\|_2 > C_1 \sigma \geq 1 - \exp(-t)/2.$$

Then we show the bound for $\|G\|_2^2$. We adapt the proof of Theorem 5.39 here. Here G is a $d \times (M+k-1)$ matrix with G_i being independent sub-Gaussian random variables in $\mathbb{R}^d$ with second moment $\Sigma = cI$. This is because:

dd

$$E[G_i^T G_j] = E[G_i^T E[G_j | \mathcal{F}_{j-M}]] = E[G_i^T] E[G_j] = 0 \text{ if } i, j \in [M],$$

$$k=1 \quad k=1$$

$$E[G_i^T G_j] = E[G_i^T G_j | \mathcal{F}_{j-M}] = 0, \text{ else } i < j,$$

where the second equality follows from $E[G_j | \mathcal{F}_{j-M}] = 0$ if $j > M$ and $G_i \in \mathcal{F}_{j-M}$ due to the martingale property.

T

$$j \in [M+k-1] \quad T$$

Then setting $\delta = C \sqrt{d} + \sqrt{t}$

$$M+k-1 \quad M+k-1$$

is equivalent to showing:

, we need to show $\| \frac{1}{T} \sum_{j=1}^T G_j - I \| \leq \max\{\delta, \delta^2\}$, which $M+k-1$

$$\|G_u\|_2^2 - 1 \leq \max\{\delta, \delta^2\} \quad 2$$

max|

for a $1/4$ -set U consisting the unit sphere S^{d-1} .

1

$$u \in U \quad M+k-1$$

To conclude the proof, denote $V = G_u$, and we only need to show: $\frac{1}{T} \sum_{i=1}^T V_i^2 - 1$

$P \left| \frac{1}{T} \sum_{i=1}^T V_i^2 - 1 \right| \geq \varepsilon \leq 2 \exp(-c_1 \min\{\varepsilon^2, \varepsilon\} T)$. Denote $S = \frac{1}{T} \sum_{i=1}^T V_i^2 - 1$, $M+k-1 \quad i=1 \quad i \leq W$

and each $V_i^2 - 1$ is sub-Exponential with parameters $4W^2$ (when $i > M$, the sub-Exponential i

property is with respect to $\mathcal{F}_{i-M}$). Then we have: $P(S \geq t) = P(e^{\lambda S} \geq e^{\lambda t}) \leq e^{-\lambda t} E[e^{\lambda S}]$

$$= e^{-\lambda t} E[\exp(\lambda$$

$$M+k-1$$

$$X(V_i^2 - 1) | \mathcal{F}_{i-M}] \leq 0 \quad V(i-M)$$

$$T(M + k - 1)$$

$$X^T B^T \phi(x_j, t, s_j, t) \phi(x_j, t, s_j, t)^T B, \forall j \in [M + k - 1], t \in [T]$$

The minimum eigenvalue of $\nabla^2 F(B, \theta)$, is lower bounded by:

where:

$$\lambda_{\min}(H) \geq \lambda_{\min} \geq$$

$$\lambda_{\phi \min} = \min_{v^T \|v\|^2=1}$$

$$2 \lambda_{\phi \min} \min(\lambda_{B \min}, \lambda_{\theta \min}), T(M + k - 1)$$

$$\|X \phi(x_j, t, s_j, t) \phi(x_j, t, s_j, t)^T v,$$

$$F(B^{\wedge}; \{\theta^{\wedge}\}) - F(B; \{\theta\}) \leq \lambda \sum_{j \in [M+k-1]} \sum_{j \in [M+k-1]}$$

max

$$(H_{M+k-1}$$

)

$$\max \|B^{\wedge} \theta - B \theta\|^2 \leq \lambda \sum_{j \in [M+k-1]} \sum_{j \in [M+k-1]}$$

max

$$(H_{M+k-1}$$

$$)E_{2M+k-1}$$

.

t,j

$$\lambda_{B^*} = \min_{w^T} w^T (B^T B) w,$$

$$\min \|w\|^2 = 1$$

$$\lambda_{\theta^*} = \min_{z^T} z^T X \theta^* \theta^{*T} z.$$

$$\min \|z\|^2 = 1 \sum_{j \in [M+k-1]}$$

$$\sum_{j \in [M+k-1]}$$

$$\sum_{j \in [M+k-1]}$$

$$\sum_{j \in [M+k-1]}$$

Following

$F(B; \{\theta_j\}_{j \in [M+k-1]})$ at the point of $B, \{\theta_j\}_{j \in [M+k-1]}$ (since F is strongly convex with respect to B and $\{\theta_j\}_{j \in [M+k-1]}$), we have:

this result,

taking a second-order Taylor

expansion of

$$F(\hat{B}; \{\hat{\theta}\}) - F(B; \{\theta\})$$

$$\geq$$

$$\lambda_{\min}(H_{M+k-1})^2$$

$$\frac{1}{2} \|\hat{B} - B\|^2 + F$$

$$\times$$

$$i \in [M+k-1]$$

$$\|\theta - \hat{\theta}\|^2 \leq \frac{1}{\lambda_{\min}(H_{M+k-1})} \sum_{i=1}^M \|\theta_i - \hat{\theta}_i\|^2$$

Then we have:

$$\max\{\|\hat{B} - B\|^2, \sum_{i=1}^M \|\theta_i - \hat{\theta}_i\|^2\} \leq \lambda_{\min}(H_{M+k-1}) = \kappa(H_{M+k-1})(EM+k-1)^2. \quad \square$$

$$\hat{\theta} \leq \lambda_{\max}(H_{M+k-1})(EM+k-1)^2$$

EC.2.2. Greedy

EC.2.2.1. Technical Lemmas In the context when no confusion arises, we use the single

index for $\{(s, x, c)\}$ to denote the state, action, and the cost; and the estimate for the $\theta, \hat{\theta}_i$ $i \in [T]$

in the learning process. All of these subscripts of notations ignore the dependence with respect to the $(M+1)$ -th batch.

ec6 e-companion to :

Proposition EC.1 (Online Estimation Error of OLS under Misspecified Representation).

For general $\hat{B} \neq B$ with Consider the data $\{(s_i, x_i, c_i)\}_{i \in [n]}$ where $\{(s_i, x_i)\}_{i \in [n]}$ are fixed, satisfying $c_i = \phi(x_i, s_i)^\top B \theta + \epsilon_i$. Denote the following regression:

$$\frac{1}{n} \sum_{i=1}^n$$

$$\hat{\theta}^n = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n (\phi(x_i, s_i)^\top \hat{B} \theta - c_i)^2,$$

$$\theta = \frac{1}{n} \sum_{i=1}^n$$

$$\theta = \arg \min_{\theta} \mathbb{E} \left\{ \frac{1}{n} \sum_{i \in [n]} (\phi(x_i, s_i)^\top B \theta - c_i)^2 \right\}, \quad \theta_i = 1$$

where $E\{\varepsilon_i | i \in [n] \} [\cdot]$ is taken randomness over the noise. Denote $Z_{B,n} \in \mathbb{R}^{n \times p}$ (and $Z_{B^{\wedge},n} \in \mathbb{R}^{n \times p}$) as the matrix whose i -th row is $(B^T \phi(x_i, s_i))$ (and $B^{\wedge T} \phi(x_i, s_i)$) respectively, $i \in [n]$.

With probability

$$1 - 12 \exp(-t):$$

Proof of Proposition EC.1. Denote θ^* as the best-in class parameter under the misspecified

representation $B^{\wedge}$, that is:

$$\frac{1}{n} \sum_{i=1}^n$$

$$\theta^* = \arg \min_{\theta} E\{\varepsilon_i | i \in [n]\} (\phi(x_i, s_i)^T B^{\wedge} \theta - c_i)^2.$$

$$\theta_i = 1$$

Denote $\Sigma = Z^T Z / n$ and $\Sigma^{\wedge} = Z^{\wedge T} Z^{\wedge} / n$. Then we bound $\|\theta^{\wedge} - \theta\| \leq \|\theta^{\wedge} - \theta^*\| + \|\theta^* - \theta\|$.

$$\sigma^2(p + 2\sqrt{pt} + 2t) \leq tCD \|B^{\wedge} - B\|_F$$

$$\|\theta^{\wedge} - \theta\|^2 \leq \lambda(Z^T Z) + \lambda(Z^{\wedge T} Z^{\wedge} / n). \quad (\text{EC.2})$$

$$\min_{B,n} B, n \min_{B,n} B, n$$

$$B, n B, n B^{\wedge}, n B, n \frac{2}{n} \frac{2}{n} \frac{2}{n}$$

The first term $\|\theta^{\wedge} - \theta^*\|^2$ is related to the variance in the estimation of $\theta^{\wedge}$, from Remark 8 in Hsu

et al. (2012a), we have: with probability at least $1 - 1 \exp(-t)$, $\|\theta^{\wedge} - \theta^*\|^2 \Sigma \leq \sigma^2(p + 2\sqrt{pt} + 2t)$. Then we

$$\text{can bound } \|\theta^{\wedge} - \theta^*\| \leq \sigma(p + 2\sqrt{pt} + 2t) \cdot \frac{1}{\lambda_{\min}(Z^T Z_{B,n})}$$

$$B, n$$

The second term $\|\theta^* - \theta\|^2$ is related to the bias in the estimation $B^{\wedge}$. By plugging in the exact formula of θ and θ^* , with probability at least $1 - 1 \exp(-t)$, we have:

$$\|\theta^* - \theta\| = (Z^T Z)^{-1} Z^T c - (Z^{\wedge T} Z^{\wedge})^{-1} Z^{\wedge T} c$$

$$\frac{2}{n}$$

$$B, n B, n B, n B^{\wedge}, n B, n B^{\wedge}, n \Sigma$$

$$\leq \|(Z^T Z)^{-1} \| Z^T - Z^{\wedge T} \| c\| + \|(Z^{\wedge T} Z^{\wedge})^{-1} \| Z^{\wedge T} Z^{\wedge} - Z^T Z \| (Z^{\wedge T} Z^{\wedge})^{-1} \| \| c\|$$

$$\leq$$

$$tCD$$

$$\lambda(ZTZ/n) \min B,n B,n$$

$$\|B^{\wedge} - B\|_F.$$

$$B,n B,n$$

$$tC$$

$$\lambda(ZTZ)/n \min B,n B,n$$

$$B,n B^{\wedge},n B,n \|(B^{\wedge} - B) \sup \phi(x_i, s_i)\|_2 \leq$$

$$i$$

$$B,n B^{\wedge},n B,n B,n B,n B^{\wedge},n B$$

where $c = (c_1, \dots, c_n)^T$ and the first inequality follows by the identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. $\square$

Then we present the following martingale inequality for matrix, which is extracted from Theorem

3.1 in Tropp (2011):

Lemma EC.6 (Matrix Martingale Inequality). Consider $F_1 \subset F_2 \dots$ be a filtration and a finite sequence of positive semi-definite matrices $\{X_k\}$ with dimension d . Suppose $\lambda_{\max}(X_k) \leq R$ almost surely. Define the summation $Y := \sum_{k=1}^n X_k$ and $W := \sum_{k=1}^n E[X_k | F_{k-1}]$. Then for all $\mu \geq 0, \gamma \in [0, 1]$, we have:

$$\mu \otimes e^{-\gamma} \otimes R$$

$P(\lambda_{\min}(Y) \leq (1-\gamma)\mu \text{ and } \lambda_{\min}(W) \geq \mu) \leq d(1-\gamma)^{1-\gamma}$. Based on that, we can show the following matrix property hold:

e-companion to : ec7

Lemma EC.7 (Minimum Eigenvalue Concentration). Recall $Z_{B,n}$ defined in Proposi

tion EC.1 and suppose Assumption 5 holds. Then with probability $1 - \delta$, for any $T_0 \leq n \leq T$,

$$\lambda(ZTZ) \geq n, \text{ where } T \geq 4D^2 \log(pT/\delta). \min B,n B,n \geq 0.1c_1$$

Proof of Lemma EC.7. We apply matrix concentration inequality in Lemma EC.6 to the sequence $\{B^T \phi(x_k, s_k)(B^T \phi(x_k, s_k))^T\}_{k \in [T]}$. From Assumption 5, we have: $\lambda_{\min}(E[B^T \phi(x_{k+1}, s_{k+1})(B^T \phi(x_{k+1}, s_{k+1}))^T | F_k]) \geq \min_x \lambda_{\min}(E[B^T \phi(x, s)(B^T \phi(x, s))^T]) \geq$

c . Thus, by setting $Y = P$

$$\frac{1}{T} \sum_{k \in [T]}$$

$$\|B^T \phi(x, s) - (B^T \phi(x, s))^T\|_F = \sqrt{2} \sqrt{\lambda(B)}$$

, $W = E[B^T \phi(x_{k+1}, s_{k+1})(B^T \phi(x_{k+1}, s_{k+1}))^T | \mathcal{F}_k]$, $\gamma = 1/2$, $R = D$, $\mu = c_1 n$ in Lemma EC.6, we have:

$$\sqrt{c_1 n} \leq \sqrt{2} \sqrt{\lambda(B)}$$

EC.2.2.2. $K = 1$. In the proof of the greedy algorithm, we directly consider the case when B is unknown while leveraging (EC.2) in Proposition EC.1. Note that when B is known, the proof follows the same while setting $B = \hat{B}$ in (EC.2). In the following, denote x^i as the action taken by Algorithm 2 and x^*_i at each time $i \in [T]$.

n

$$\leq 1 \leq p \leq 2e^{-0.5 \sqrt{D} \sqrt{n}}$$

$$0.1cn$$

1).

$$P(\lambda(Z^T Z) \leq \min_{B, n}$$

$$= \exp(-\log p -$$

Applying the above with a union bound for every $T_0 \leq n \leq T$, with probability at least $1 - \delta$, we

T

B, n

$$4D^2$$

have: $\lambda(Z^T Z) \geq c_1 n$ for any $n \in [T, T]$. $\square$

$$\min_{B, n} \sqrt{\lambda(B, n)} \geq 0$$

T

$$E[R_1, T] = \sum_{i=1}^T \phi(x^i, s_i)^T B \theta - \phi(x^*_i, s_i)^T B \theta$$

$$i=1 \quad T$$

$$= \sum_{i=1}^T \phi(x^i, s_i)^T B \theta - \phi(x^i, s_i)^T \hat{B} \theta + \sum_{i=1}^T \phi(x^*_i, s_i)^T \hat{B} \theta - \phi(x^*_i, s_i)^T B \theta$$

(EC.3) (EC.4)

$$i=1$$

$$+ \sum_{i=1}^T \phi(x^i, s_i)^T \hat{B} \theta - \phi(x^*_i, s_i)^T \hat{B} \theta,$$

$$i=1$$

where the error in (EC.3) arises due to the representation error and becomes zero when $B = \hat{B}$. The error in (EC.4) arises due to the estimation error from $\hat{\theta}$.

We bound them one by one. First, consider the following event:

$A =$

$$\lambda_{\min}(Z_B, n Z_B, n) \geq \frac{1}{2},$$

$$\forall T_0 \leq n \leq T,$$

.

$$\|\hat{\theta} - \theta\| \leq \frac{1}{\sqrt{n}} \left(\sqrt{p \log T + 2 \log T} + C D \log T \|\hat{B} - B\|_F \right), \quad \forall T_0 \leq n \leq T,$$

$$E \leq d \log T, \quad M \leq M \alpha T$$

$$\kappa(H, M) \leq \kappa < \infty$$

$$\frac{1}{2} \leq \frac{1}{2}$$

Within A , based on Proposition EC.1 for the first part, Lemma EC.7 for the second and fourth part, Proposition 1 for the third part, we know: $P(A^c) \leq C/T$ for some constant C .

Under the event of A , for the term in (EC.3), following Proposition 2, we have:

$\mathbb{E}$

$$\mathbb{E} \left[\sum_{i=1}^n \phi(x_i, s_i)^\top \hat{\theta} - \phi(x_i, s_i)^\top \theta \right] + \mathbb{E} \left[\sum_{i=1}^n \phi(x_i, s_i)^\top \hat{\theta} - \phi(x_i, s_i)^\top \theta \right]$$

$$\leq \sum_{i=1}^n \|\hat{\theta} - \theta\|_2 \|\phi(x_i, s_i)\|_2 \leq \sqrt{n} \|\hat{\theta} - \theta\|_2 \max_{i \leq n} \|\phi(x_i, s_i)\|_2$$

$M \alpha$

x, s

$$\leq 2D \|\hat{\theta} - \theta\|_2 \sqrt{n} \leq 2D \|\hat{\theta} - \theta\|_2 \sqrt{n} M \alpha.$$

$\sqrt{\cdot}$

$i=1$

$r \, dT$

$$h(T) = T^{\frac{1}{2}} \left(\frac{1}{2} \log T + \frac{1}{2} \log \frac{1}{2} \right)$$

$$\mathbb{E} \sum_{i=1}^n \phi(x_i, s_i)^\top \hat{\theta} - \phi(x_i, s_i)^\top \theta + \mathbb{E} \sum_{i=1}^n \phi(x_i, s_i)^\top \hat{\theta} - \phi(x_i, s_i)^\top \theta \leq 2D \|\hat{\theta} - \theta\|_2 \sqrt{n} dT + D.$$

ec8 e-companion to : For the term in (EC.4), for each $i \in [T]$, we have:

$$\phi(\hat{x}, s)TB^{\hat{\theta}} - \phi(x^*, s)TB^{\hat{\theta}} \leq \phi(\hat{x}, s)TB^{\hat{\theta}} - \phi(\hat{x}, s)TB^{\theta^*} + \theta^*TB^{\hat{\theta}}\phi(x^*, s) - \theta TB^{\hat{\theta}}\phi(x^*, s)$$

$$\sigma\sqrt{p(\log T)^{1/2}XT r 1 \kappa CDE !} \leq 4D2 \sqrt{\cdot} + T M$$

$$\leq T\|\theta^{\wedge} - \theta\| \|B^{\wedge}\| (\|\phi(\hat{x}, s)\| + \|\phi(x^*, s)\|) i2Fi2ii2$$

$$\leq 2D\|\theta^{\wedge} - \theta\| \|B\| + \sqrt{\kappa}E i2FM$$

$$\leq 4 D 2 \|\theta^{\wedge} - \theta\|, i2$$

where the first inequality follows from the optimality of the Greedy algorithm, that is: $\phi(\hat{x}, s)TB^{\hat{\theta}} \leq \phi(x^*, s)TB^{\hat{\theta}}, \forall i \in [T]$ and the last inequality holds since $p_{\kappa}(H)E \leq \|B\|$ when

$$T \geq T_0.$$

Summing them from 1 to T , we have:

$$\sum_{i=1}^T E X \phi(\hat{x}^i, s_i)TB^{\hat{\theta}} - \phi(x^*, s_i)TB^{\hat{\theta}}$$

$$\leq E 4D2X\|\theta^{\wedge} - \theta\| i2$$

$$i=1$$

$$\sqrt{T} \sqrt{s}$$

$$\sqrt{\cdot}$$

$$\leq E 4D2 X \sigma^2(p+2 p\log T+2\log T)+CD\log T\|B^{\wedge}-B\|F c1n c1$$

$$n=T_0$$

$$LEm + (P(A)2DT+T_0) \|B\|F+ \zeta$$

$$\sqrt{T} \sqrt{s} \sqrt{\cdot}$$

$$\leq E 4D\|B\| X \sigma^2(p+2 p\log T+2\log T) +CD\log T\|B^{\wedge}-B\|F |A +\Theta(1)$$

$$F c1n c1 n=T_0$$

$$c1 n=T n c1 0$$

$$\sqrt{pT\log T 4pD2C rdT \sqrt{\cdot}} 11$$

$$rd!\sqrt{\cdot} \leq 4D2\sqrt{c+\alpha} M+T=\Theta 1+MapT,$$

The third inequality follows from Assumption 5 and Lemma EC.7 such that $P(A) \geq 1 - C/T$.

The last equality ignores the constant with respect to D and α .

Therefore, by combining the result in (EC.3) and (EC.4), we have:

$$\mathbb{E}[R_1, T] \leq O(\sqrt{1 + M} \sqrt{pT}).$$

As a special case when B is known, $\hat{B} = B$, we repeat the previous arguments while eliminating the representation error (e.g., Equation (EC.3) becomes 0). This leads to the error bound of $O(\sqrt{pT})$. $\square$

EC.2.2.3. $K > 1$. When B is unknown, following the same analysis as in Appendix EC.2.2.2, for the k -th “test” batch, the expected regret is upper bounded by $(1 + qd)\sqrt{pT}$.

$$K \sqrt{d} \sqrt{1 + qd} \sqrt{pT}$$

$$\alpha(M + k - 1)$$

Denote the regret at each new online round k as R_k . Following, we sum the total regret across

K batches. This gives: $\mathbb{E}[R_K, T] =$

$$1 + \alpha(M + k - 1) \sqrt{pT}$$

e-companion to :

ec9

$$\sqrt{rd} \sqrt{K} \sqrt{1 + qd} \sqrt{pT} = K \sqrt{pT} + p \alpha \sqrt{1 + M - 1} \sqrt{T} \leq K \sqrt{pT} + p M \alpha,$$

$$\leq 2\sqrt{d} \sqrt{q} \sqrt{1 + K - 1}, \text{ and } M$$

$\square$

where the second equality follows from $\sum_{k=1}^K \sqrt{dt} \leq \sqrt{K} \sqrt{dt} \leq \sqrt{K} \sqrt{M} \sqrt{1 + t}$

$$\sqrt{1 + M}$$

the first inequality follows $1 + a \leq 1 + a$ for any $a > 0$.

When B is known, the sum of regret $\mathbb{E}[R_K, T] = K \mathbb{E}[R_1, T] \leq pK$

EC.2.3. Thompson Sampling

EC.2.3.1. Technical Lemmas We directly consider the case when B is unknown. When B

is known, the result directly follows by replacing $\mathbb{E}[R_{M+k-1}]$ with the standard error in regression for $\|\hat{\theta} - \theta\|$.

Definition EC.1 (Generalized n-Monte Carlo). We say a family of Bayesian contextual algorithms $\{\text{alg}(w) : w \in W\}$ satisfy the generalized n-Monte Carlo property if, for all possible trajectories τ_h ,

$$\text{TV}(\text{Palg}(\theta)(a_h|\tau_{h-1}) \parallel \text{Palg}(\theta')(a_h|\tau_{h-1})) \leq \text{TV}(P_{\theta}(\phi|\tau_{h-1}) \parallel P_{\theta'}(\phi|\tau_{h-1})).$$

Definition EC.2 (Tail Expectation). Let X be a nonnegative random variable on a probability space $(\Omega, \mathcal{F})$ with law P and finite expectation $E[X] < \infty$. We define its tail expectation, as a function of probabilities $p \in (0, 1]$ as:

$$\Psi_X(p) := p \sup\{E[XY], \text{ s.t. } Y : (\Omega, \mathcal{F}) \rightarrow [0, 1], \text{ and } E[Y] \leq p\}.$$

For $p > 1$, we extend $\Psi_X(p) = E[X]$. Overloading notation, we let $\Psi_w(p)$ denote the upper tail function over $\text{diam}(\mu)$ when drawn from P_w :

$$\Psi_w(p) := \Psi_{\text{diam}(\mu)}(p) \text{ where } \mu \sim P_w.$$

We extract Theorem E.1 in Simchowit et al. (2021) (with a bandit special example) as follow,

where we denote $\text{alg}(w)$ as the algorithm with prior w :

Lemma EC.8 (Cost Difference under Different Prior). Consider $\text{alg}(w)$ satisfy the nMonte Carlo property on the time period T with $\{(s_t, x_t(w), c_t(w))\}_{t \in [T]}$ as the state, action and costs generated by $\text{alg}(w)$ across the time horizon T . Denote $R_w(\text{alg}(w'))$ as the total expected cost for the algorithm with prior w' on the whole time horizon T where the true prior of the unknown parameter is w' . Then for $\varepsilon = \text{TV}(P_w, P_{w'})$:

$$|R_w(\text{alg}(w)) - R_w(\text{alg}(w'))| \leq 4nT^2\varepsilon \cdot \Psi_{\bar{w}'}(2T\varepsilon), \text{ where } \Psi_{\bar{w}'}(p) = \Psi_{|c_{T,w}|}(p) \text{ with } c_{T,w} := T^{-1} \sum_{t=1}^T c_t(\text{alg}(w)).$$

Proof of Lemma 2. We know Algorithm 3 is a 1-Monte Carlo algorithm from Simchowit et al. (2021). Then plugging the result of Lemma EC.8 in into our problem instance, we have:

$$R_{1,T}(w) - R_{1,T}(w') \leq 4T^2\varepsilon \Psi_{\bar{w}'}(2T\varepsilon) \leq 4T^2\varepsilon (\|\phi(x, s)\|_{\infty} B_{\theta} + \sqrt{\sigma}) \leq CT^2\varepsilon, \quad T$$

where the second inequality follows from Lemma B.6 in Simchowit et al. (2021) and: $\frac{1}{T} \sum_{t=1}^T$

$\frac{1}{T} \sum_{t=1}^T$

$$|c_{T,w}| = \frac{1}{T} \sum_{t=1}^T c_t(x_t, s_t) \leq \|\phi(x, s)\|_{\infty} B_{\theta} + \frac{1}{T} \sum_{t=1}^T \varepsilon_t.$$

ε_t is sub-Gaussian with the parameter $\sqrt{\sigma}$, that with probability $1 - e^{-t}$ conditioned

ε_t is sub-Gaussian with the parameter $\sqrt{\sigma}$, that with probability $1 - e^{-t}$ conditioned

result. $\square$

Since 1 PT

$T \ t=1$

-

on the randomness of $\{\epsilon_t\}_{t \in [T]}$, $\Psi_{w'}(2T \epsilon) \leq \|\phi(x, s)\|_{\infty} B \theta + \sqrt{T}$. Integrating over t yields the above

ec10 e-companion to : Lemma EC.9 (TV Distance Bound). For two Gaussian distributions $N(\mu_1, \Sigma_1)$ and

$N(\mu_2, \Sigma_2)$ supported in $\mathbb{R}^d$, then:

$$TV(N(\mu, \Sigma), N(\mu', \Sigma')) \leq \frac{1}{2} \text{tr}(\Sigma^{-1} \Sigma' - I) + (\mu - \mu')^T \Sigma^{-1} (\mu - \mu') - \log \det(\Sigma \Sigma'^{-1}).$$

11222121211221 (EC.5)

$$\leq C(\|\mu_1 - \mu_2\|_2 + \|\Sigma_1 - \Sigma_2\|_{\text{op}})$$

EC.2.3.2. $K = 1$. To prove the regret of Algorithm 1 using the component of Algorithm 3, from Lemma 2, we only need to bound the prior difference for each k . Firstly, we consider $k = 1$ and $\hat{\mu}, \hat{\Sigma}$ then denote the empirical mean and covariance of parameters in the M batches and $\hat{B} = \hat{B}^0$.

Consider P_w as the true Gaussian distribution $N(B\mu, B\Sigma B^T)$ with parameters B, μ, Σ ; $P_{\hat{w}}$ as the estimated Gaussian distribution $N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^T)$ estimated prior with parameters $\hat{B}, \hat{\mu}, \hat{\Sigma}$. We bound the misspecification error between the true prior $P_w = N(B\mu, B\Sigma B^T)$ and the estimated prior $P_{\hat{w}} = N(\hat{B}\hat{\mu}, \hat{B}\hat{\Sigma}\hat{B}^T)$. From Lemma EC.9, we only need to control the mean and covariance difference.

For the mean difference, with probability $1 - e^{-t}$ with respect to all the randomization, we have: $\|\hat{B}\hat{\mu} - B\mu\| = \|\hat{B}^T X \hat{\theta} - B^T X \theta + B^T X \theta - B\mu\|$

$$2M \sum_{i \in [M]} \sum_{j \in [M]} \sum_{k \in [M]} \|\hat{B}^T X \hat{\theta} - B^T X \theta + B^T X \theta - B\mu\|$$

$$\leq \|\hat{B}^T X \hat{\theta} - B^T X \theta\| + \|\hat{B}^T X \theta - B^T X \theta\| \sum_{i \in [M]} \sum_{j \in [M]} \sum_{k \in [M]} \|\hat{\theta} - \theta\|$$

$$\sum_{i \in [M]} \sum_{j \in [M]} \sum_{k \in [M]} \|\hat{\theta} - \theta\|$$

$$1X \ 1X \leq \|\hat{B} - B\| \|\hat{\theta}\| + \|\hat{B}\| \sum_{i \in [M]} \sum_{j \in [M]} \sum_{k \in [M]} \|\hat{\theta} - \theta\| + \|\mu - \theta\|$$

$$FM \sum_{i \in [M]} \sum_{j \in [M]} \sum_{k \in [M]} \|\hat{\theta} - \theta\|$$

$$\text{rdp!rd} \leq \|\hat{B}\| F_p M \sqrt{T} + \sqrt{M} \leq \|\hat{B}\| F_p M \sqrt{T},$$

where the third inequality follows from the fact both $\|B^\wedge - B\|_F, \max_{i \in [M+k-1]} \|\theta^\wedge_i - \theta_i\| \leq \sqrt{\frac{p}{T}}$

By using Proposition 2 and $\|P(\theta_i - \mu)\|_2 \leq \sqrt{p}$. And the last inequality follows

$M \in [M]$

For the covariance difference, similarly, we can bound:

$M \in [M]$

M

from Assumption 4.

$$\|B^\wedge \Sigma^\wedge B^\wedge - B \Sigma B^T\| \leq \|B\|_F \|\Sigma^\wedge - \Sigma\|_F + \|B^\wedge - B\|_F \|\Sigma\|_F,$$

where $\|\Sigma^\wedge - \Sigma\|_F$ is further bounded as follows:

$$\| \Sigma^\wedge - \Sigma \|_F \leq \sqrt{p} + \sqrt{p}.$$

$$\| \Sigma^\wedge - \Sigma \|_F \leq \sqrt{p} + \sqrt{p}.$$

$M \in [M]$

Combining them together, we have:

$$R_{1,T}(w') = R_{1,T}(w) + R_{1,T}(w') - R_{1,T}(w)$$

$$\sqrt{2} \sqrt{pT} + \sqrt{pT} \leq pT + T \cdot \text{TV}(N(B\mu^\wedge, B\Sigma B^T), N(B\mu, B\Sigma B^T)) \leq pT + pM.$$

where the second inequality follows from the regret of Thompson sampling under the true prior is

$1 \leq M$

$$\| \Sigma^\wedge - \Sigma \|_F \leq \sqrt{p} + \sqrt{p}.$$

$$\| \Sigma^\wedge - \Sigma \|_F \leq \sqrt{p} + \sqrt{p}.$$

$O(p)$

$\sqrt{p}$

T) from Russo and Van Roy (2014).

$$\| \theta^\wedge_i - \theta_i \|_2 +$$

$$\| \theta^\wedge_i - \theta_i \|_2 +$$

ec11 EC.2.3.3. $K > 1$. Following the The expected k -th round regret is bounded by $pT +$

$\sum_{k=1}^K \mathbb{E}[\text{RK}, T] = (M + K - 1)\alpha$

e-companion to :

$$\mathbb{E}[\text{PK}] = \mathbb{E}[\sum_{k=1}^K p\sqrt{T} + pq \text{ dT3}] = \mathbb{E}[\sum_{k=1}^K pK\sqrt{T} + qdKT3] .$$

$$k=1$$

EC.3.

EC.3.1.

We discussed the data extraction from procedure different sources in the dataset, with details referred to Martagan et al. (2024). For the state variables, we describe data from the following two sources:

Temperature and Humidity: We include the temperature and humidity from the records, which are measured every 15 minutes.

Raw Material: We consider sum raw material attributes, especially those related to raw materials that vary across batches of new material. Specifically, we focus on the amount of API, the amount of Magnesium Stearate, and the lot (bag) of Magnesium Stearate. The amount of API and Magnesium are included since they change with each refill, and the lot (bag) of Magnesium Stearate is also considered due to its limited yet meaningful variability (4 different lots across 57 refills). Although the refill frequencies between Magnesium Stearate and API differ slightly, we align them according to the refill time of API. As mentioned in the main body, the time horizon between consecutive API refills typically last 12 hours.

Note that there are two different lots of Avicel PH 102. However, one lot is used predominantly with Avicel PH 102 being fed in six consecutive hours, while most of the raw materials are fed from the other lot at different times, resulting in trivial data imbalance. Besides, we do not observe significant differences between these two lots in terms of the action and cost performance from Section 3.

As described in Figure 2, these state variables are used in both blending and compression stages.

For the blending stage, we extract the actions and the cost from the following data sources:
LiW-Feeder Data: We extract the LiW feeder screw speed as the actions to adjust, which controls

the corresponding massflow of each feeder. Each entry is measured every second.

Blender Data: We set the cost of the screw speed as the summation of total loss in Table 1, i.e.,

for each time:

Cost = X Massflow Blender i $\times$ ($|\text{Potency Blender } i - 100| + \text{OOS Concentration } i$). (EC.6)

$i \in \{1, 2\}$

This controls the total quantity loss multiplied by their quality. Each entry is measured every second.

Therefore, for each batch i , the total time length of the dataset $\{(s_i, t, x_i, t, c_i, t)\}_{t \in [T]}$ is 12 hours, with each t partitioned by second. We filter the dataset by dropping the values where sum feeder(s) do not work, i.e., when some coordinates of x_i, t is non-positive.

For the compression stage, we extract the actions and the cost from the following data sources: Tablet Press: We choose the precompression force, main compression force, ejection force tablet, and compression cycle fill depth as four actions since they usually control the quality attribute of

the tablet. Each entry is measured every second.

Tablet Properties: We set the cost of the compression steps as the absolute deviation of average

hardness against the optimal quality in the golden profile, 14.5 (Holman et al. 2021). We do not choose other variables since other quality attributes, i.e., weight and thickness, are of less variability. Each entry is averaged over ten results, around 15 minutes.

Finally, we normalize the variables in each stage by scaling each variable of (s, x, c) in each dataset by its mean and standard deviation.

$(M + k - 1) \alpha M \alpha$

Additional Experimental Details in Section 6 Data Selection and Preprocessing

ec12 e-companion to :

EC.3.2. Experimental Details in Section 6.2

We measure the performance of the poly and forest model across each training criterion via tuning the following hyperparameters:

- MTL-Forest and Pool-Forest: tree number d in $\{5, 10, 20\}$, max depth of each tree in $\{2, 3, 5\}$;
- Ind-Forest: tree number d in $\{2, 3, 5\}$, max depth of each tree in $\{3, 5\}$;
- MTL-Poly: the representation dimension $p \in \{5, 10, 20\}$, poly feature degree in $\{1, 2, 3\}$;
- Pool-Poly and Ind-Poly: poly feature degree in $\{1, 2, 3\}$ since there are no representation B

involved.

In each forest model, we fix the max feature as 0.5, that is, we use 50% of all the features randomly for each tree.

EC.3.3. Experimental Details in Section 6.3

We detail the optimization steps here. Note that across Algorithms 2 and 3, at each time period, we are facing the following optimization problem: Given each state s , we obtain the optimal action by:

$$a^* \in \arg\min_{\phi(s, x)} TB\theta. \quad (\text{EC.7}) \quad x \in X$$

In X , we set each component of the decision as follows:

$$X = \{x \in \mathbb{R}^{dx} \mid L_i \leq x_i \leq U_i, i \in [dx]\}. \quad (\text{EC.8})$$

For $i \in [dx]$, we choose 40% and 60% quantile across the decision range for L_i and U_i to eliminate the effects of extreme values. which is a reasonable decision boundary that does not deviate from historic policy and has the potential to outperform existing benchmarks. This also reflects the reality that we do not want extreme actions that may lead to changes in temperature or humidity.

For each optimization problem in (EC.7), given the complexity of the objective function – whether modeled by a random forest (or a third-order polynomial function), we estimate the solution using a random search approach. Specifically, we randomly select 1000 points within the feasible region defined by (EC.8) and evaluate the best one. The performance and their observed trends do not change when varying the number of search points, demonstrating the robustness of this random search procedure.

EC.3.4. Additional Results in Section 6.3

We report the improvement over the history policy for greedy and TS algorithms with different random splits for the random forest model in Figure EC.3. In most cases, greedy

and TS algorithms achieve significantly better performance than the history policy, around 20% improvements.

6 4 2 0

-0.2

0.0 0.2

Cost Improvements over History Policies across Batches
greedy, pval = 0.8 TS, pval = 0.520

Cost Improvements over History Policies across Batches

6

4

2 0 0

-0.6

-0.4 -0.2 0.0 0.2 0.4 0.6

Cost Improvement (Relative)

(a) Blending 1

Cost Improvements over History Policies across Batches greedy, pval = 0.000

TS, pval = 0.000

0.4 0.6 0.8

-0.25

0.00 0.25

0.50 0.75

Cost Improvement (Relative)

(e) Compression 1

(f) Compression 2

6 4 2

Cost Improvement (Relative)

(b) Blending 2

(c) Blending 3

(g) Compression 3

(d) Blending 4

(h) Compression 4

Figure EC.3 Improvements over the history policy under the multi-task random forest over other random splits.